

CO2 - AT1 - Database Design and Connectivity

NAME: VISHWA T

REG NO: 192424180

COURSE NAME: DSA0503 - Query Processing for Data Science

Question 2: Examination Result Management System:

Aim:

To create Student and Result tables, insert examination records, calculate the total marks of each student using SQL queries, and display the results.

Algorithm:

1. · Import the sqlite3 and pandas libraries.
2. · Create a connection to the SQLite database.
3. · Create the **Supplier** table with Supplier_ID as the primary key.
4. · Create the **Product** table with Product_ID as the primary key and Supplier_ID as the foreign key.
5. · Insert sample records into the **Supplier** table.
6. · Insert sample records into the **Product** table.
7. · Update the stock quantity of a selected product.
8. · Execute an SQL **JOIN** query to retrieve product details along with supplier information.
9. · Display the retrieved records.
10. · Close the database connection.

CODE:

```
import sqlite3
```

```
import pandas as pd
```

```
conn = sqlite3.connect("InventoryDB.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Supplier(
```

```
    Supplier_ID INTEGER PRIMARY KEY,
```

```
    Supplier_Name TEXT NOT NULL,
```

```
    City TEXT
```

```
)
```

```
""")
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Product(
```

```
    Product_ID INTEGER PRIMARY KEY,
```

```
    Product_Name TEXT NOT NULL,
```

```
    Stock_Quantity INTEGER,
```

```
    Price REAL,
```

```
    Supplier_ID INTEGER,
```

```
    FOREIGN KEY(Supplier_ID) REFERENCES Supplier(Supplier_ID)
```

```
)
```

```
""")
```

```
cursor.execute("DELETE FROM Product")
```

```
cursor.execute("DELETE FROM Supplier")
```

```
suppliers = [  
    (1, "ABC Traders", "Chennai"),  
    (2, "XYZ Suppliers", "Bangalore"),  
    (3, "Global Mart", "Hyderabad")  
]
```

```
cursor.executemany("INSERT INTO Supplier VALUES(?,?,?)",  
suppliers)
```

```
products = [  
    (101, "Laptop", 20, 55000, 1),  
    (102, "Mouse", 50, 500, 2),  
    (103, "Keyboard", 30, 1200, 3)  
]
```

```
cursor.executemany("INSERT INTO Product VALUES(?,?,?,?,?)",  
products)
```

```
cursor.execute("UPDATE Product SET Stock_Quantity = 40 WHERE  
Product_ID = 101")
```

```
conn.commit()
```

```
query = ""
```

```

SELECT Product.Product_ID,

       Product.Product_Name,

       Product.Stock_Quantity,

       Product.Price,

       Supplier.Supplier_Name,

       Supplier.City

FROM Product

INNER JOIN Supplier

ON Product.Supplier_ID = Supplier.Supplier_ID

"""

```

```
df = pd.read_sql_query(query, conn)
```

```
print("Inventory and Supplier Details\n")
```

```
print(df)
```

```
conn.close()
```

OUTPUT:

```

..  Inventory and Supplier Details

   Product_ID Product_Name  Stock_Quantity  Price  Supplier_Name  City
0         101      Laptop             40  55000.0    ABC Traders  Chennai
1         102       Mouse             50   500.0    XYZ Suppliers  Bangalore
2         103   Keyboard             30  1200.0    Global Mart  Hyderabad

```

Question 2: Examination Result Management System

Aim:

To create Student and Result tables, insert examination records, calculate the total marks of each student using SQL queries, and display the results.

Algorithm:

- 1) Import the SQLite and Pandas libraries.
- 2) Create a database connection.
- 3) Create the Student table.
- 4) Create the Result table with a foreign key referencing Student.
- 5) Insert student records.
- 6) Insert examination result records.
- 7) Calculate total marks using an SQL query.
- 8) Display the student results.
- 9) Close the database connection.

CODE:

```
import sqlite3

import pandas as pd

conn = sqlite3.connect("ExamDB.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student(

    Student_ID INTEGER PRIMARY KEY,

    Student_Name TEXT NOT NULL,

    Department TEXT
```

)

""")

cursor.execute("""

CREATE TABLE IF NOT EXISTS Result(

Result_ID INTEGER PRIMARY KEY,

Student_ID INTEGER,

Subject1 INTEGER,

Subject2 INTEGER,

Subject3 INTEGER,

FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID)

)

""")

cursor.execute("DELETE FROM Result")

cursor.execute("DELETE FROM Student")

students = [

(101, "Arun", "CSE"),

(102, "Priya", "AI&DS"),

(103, "Rahul", "ECE")

]

```
cursor.executemany("INSERT INTO Student VALUES(?,?,?)", students)
```

```
results = [  
    (1, 101, 85, 90, 88),  
    (2, 102, 92, 87, 95),  
    (3, 103, 78, 80, 82)  
]
```

```
cursor.executemany("INSERT INTO Result VALUES(?,?,?,?,?)", results)
```

```
conn.commit()
```

```
query = """
```

```
SELECT Student.Student_ID,
```

```
    Student.Student_Name,
```

```
    Student.Department,
```

```
    Result.Subject1,
```

```
    Result.Subject2,
```

```
    Result.Subject3,
```

```
    (Result.Subject1 + Result.Subject2 + Result.Subject3) AS  
Total_Marks
```

```
FROM Student
```

```
INNER JOIN Result
```

```
ON Student.Student_ID = Result.Student_ID
```

```
"""
```

```
df = pd.read_sql_query(query, conn)
```

```
print("Examination Results\n")
```

```
print(df)
```

```
conn.close()
```

OUTPUT:

```
*** Examination Results
```

	Student_ID	Student_Name	Department	Subject1	Subject2	Subject3	\
0	101	Arun	CSE	85	90	88	
1	102	Priya	AI&DS	92	87	95	
2	103	Rahul	ECE	78	80	82	

	Total_Marks
0	263
1	274
2	240